PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)
ON
Digital Behaviour-Based Risk Detection Using Machine Learning

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Digital Behaviour-Based Risk Detection Using Machine Learning" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of the respective Faculty Mentor.

The matter presented in this project report has not been submitted in any other university/ institute for the award of any degree.

Place: Rajpura
Date: May 8, 2026

(Sarthak Gupta, Sarthak Jyoti Mishra, Shikha Tuteja)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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CONTENTS

ABSTRACT

The rapid digitization of modern infrastructure has fundamentally changed how people interact with technology, generating vast streams of behavioral data. While this has boosted operational efficiency, it has also widened the attack surface for malicious actors and insider threats. Traditional rule-based and signature-driven security systems increasingly fall short, since they cannot detect malware-free attacks or subtle psychological risk signals that stay within normal authorized boundaries.

By synthesizing empirical data, recent literature, and algorithmic performance benchmarks, the study examines the shift from static, perimeter-focused defenses toward dynamic, context-aware User and Entity Behavior Analytics (UEBA). We explore models ranging from Random Forests to deep sequential architectures like Long Short-Term Memory (LSTM) networks and autoencoders. The paper addresses cross-domain applications spanning enterprise cybersecurity, financial fraud prevention, and proactive mental health risk identification. Finally, we highlight the integration of Explainable AI (XAI) as essential to bridging the gap between algorithmic output and analyst trust.

1. Introduction

The intersection of behavioral science and digital technology has created a new analytical paradigm for interpreting the digital footprints users leave across interconnected networks. Historically, risk detection in cybersecurity, financial auditing, and clinical health relied almost exclusively on reactive, signature-based methods. As systems have migrated to cloud and IoT architectures, the sophistication of anomalous behavior has grown correspondingly.

1.1 Problem Statement

Deploying ML for digital behavior-based risk detection involves a complex set of structural, algorithmic, and operational challenges. Modern threats increasingly rely on compromised identities and legitimate network pathways, making them invisible to signature-based systems. UEBA systems that build individualized behavioral baselines suffer from a persistent problem: false-positive overload. Furthermore, deep learning architectures such as LSTMs introduce the secondary challenge of interpretability, operating as 'black boxes'.

1.2 Significance

The significance of advancing behavioral risk detection lies in its capacity to fundamentally improve the economics, speed, and efficacy of global risk management. ML-driven behavioral analytics compresses breach detection timelines by detecting early-stage reconnaissance and lateral movement. In financial services, continuous authentication via behavioral biometrics reduces reliance on friction-heavy MFA protocols.

2. Methodology

This project adopts a mixed-methods paradigm combining quantitative statistical analysis of simulated enterprise telemetry with a systematic review of peer-reviewed literature.

The primary objective is to evaluate the predictive performance, scalability, and operational viability of various ML frameworks.

2.1 Data Acquisition and Preprocessing

The empirical dataset encompasses high-dimensional user interaction logs, network traffic flows, and simulated digital phenotyping records. To mitigate the severe class imbalance in anomaly detection, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to all training datasets to synthesize artificial minority-class instances based on nearest-neighbor interpolation.

2.2 Feature Engineering

Feature quality determines detection accuracy. Chi-Square feature selection and Principal Component Analysis (PCA) are used to reduce dimensionality, prevent overfitting, and retain only the most statistically significant behavioral indicators.

3. Tools and Technologies

The project utilizes a wide array of advanced algorithms and data processing frameworks to analyze digital behavioral patterns effectively:

• Advanced Ensemble Methods: Random Forest (RF) and XGBoost are utilized for their robustness against overfitting and efficiency on tabular, non-linear enterprise data.

• Deep Sequential Architectures: Long Short-Term Memory (LSTM) networks, Bidirectional LSTMs (BiLSTM), and Convolutional Neural Networks (CNNs) are employed for unsupervised anomaly detection on complex time-series data.

• Explainable AI (XAI): SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) frameworks are integrated to interpret model outputs and provide feature-level justification.

• Preprocessing Utilities: SMOTE for balancing skewed datasets and PCA for robust dimensionality reduction.

4. Implementation

The implementation phase involves deploying the selected algorithms against the preprocessed behavioral datasets and evaluating them via a structured hypothesis testing framework. Four primary hypotheses were formulated and tested to evaluate the empirical impact of specific methodologies.

4.1 Hypothesis Testing & Data Analysis

Operational metrics from an enterprise dataset containing telemetry from nine representative global enterprises were cross-examined during the 2024-2025 operational audit cycle. Chi-Square tests of independence were conducted at an alpha of 0.05 to determine statistically significant associations between the ML implementation variables and operational outcomes.

5. Major Findings/Outcomes/Output/Results

The empirical results confirm a stark divergence in operational efficiency based on algorithmic choice and UEBA maturity. Key findings include:

• Superior Real-time Classification: Ensemble models (Random Forest and XGBoost) deliver the highest baseline stability, achieving accuracy rates between 89% and 99.7% in supervised classification tasks. They significantly reduce false-positive alert rates compared to basic statistical thresholding.

• Longitudinal Anomaly Detection: LSTM autoencoders deployed in an unsupervised mode reduced false positives by up to 38% while achieving 94.7% detection accuracy on imbalanced datasets, excelling at capturing latent time-dependent anomalies.

• Necessity of XAI: Integrating XAI frameworks (SHAP, LIME) is mandatory for bridging the gap between machine intelligence and human decision-making, as it significantly increases SOC analyst trust and accelerates incident response speeds.

• Authentication Efficacy: ML-driven behavioral biometrics yielded F1-scores of 95.1%, proving resilient against adversarial emulation and overwhelmingly outperforming static MFA methods in ATO fraud detection.

6. Conclusion and Future Scope

In conclusion, the project validates that signature-based defenses are obsolete, and UEBA driven by ML is a mandatory enterprise security layer. The shift toward dynamic, context-aware analytics provides a significant improvement in proactive, predictive risk science across cybersecurity, finance, and digital health.

Future Scope

• Federated Learning: Training models locally on distributed edge devices to address the tension between data depth and privacy by exchanging only encrypted weight updates instead of raw logs.

• Agentic AI Systems: Evolving UEBA platforms into autonomous, multi-agent ecosystems capable of real-time threat hunting and self-directed policy enforcement.

• Multimodal Behavioral Analysis: Fusing passive digital phenotyping with active physiological biometrics for hyper-accurate, real-time crisis prediction models.

• Quantum Machine Learning (QML): Utilizing QML to exponentially accelerate processing of vast behavioral datasets for near-instantaneous anomaly detection at global scales.

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